# WARS, DEBT, AND TAXES IN THE U.S.

Fiscal and Monetary Policy 2023

Piotr Żoch

March 25, 2025

# PLAN

- We look at how the U.S. financed its wars over the last 200 years.
- We confront these patterns with the predictions of two theories: Barro (1979) and Lucas and Stokey (1983).
- We follow the chapter of the Handbook of Historical Economics by Hall and Sargent (2021) – but we omit many interesting institutional details.

# TWO THEORIES

# TWO THEORIES

- Recall the two theories we discussed earlier:
  - Barro (1979): incomplete markets
  - Lucas and Stokey (1983): complete markets
- These two theories have different recommendations regarding the optimal mix of taxes and debt to finance wars.

## TWO THEORIES

- We will consider a simplified version of Lucas and Stokey (1983) – we studied it in week 1
- In both theories the government sets taxes and issues debt to minimize deadweight loss of taxation

$$\min_{\{\tau_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \left[ \left( \frac{1}{1+r} \right)^t D\left(\tau_t\right) \right]$$

- We assume government expenditures $g_t$ are a function of a Markov state $s_t$, $g_t = G\left(s_t\right)$. Let

$$s_{t+1} \sim \phi\left(s_{t+1} \mid s_t\right)$$

where $\phi\left(s_{t+1} \mid s_t\right)$ is a Markov transition probability distribution defined on the time invariant state-space.

# TWO THEORIES

- Let $PVG_t$ be the expected present value of future government expenditures:

$$PVG_t = \mathbb{E}\left[\sum_{j=0}^{\infty}\left(\frac{1}{1+r}\right)^j g_{t+j} \mid s_t\right] = H\left(s_t\right).$$

- $H$ satisfies the functional equation

$$\begin{aligned} H\left(s_t\right) &= G\left(s_t\right) + \frac{1}{1+r}\int H\left(s_{t+1}\right)\phi\left(s_{t+1} \mid s_t\right) ds_{t+1} \\ &= G\left(s_t\right) + \frac{1}{1+r}\mathbb{E}\left[H\left(s_{t+1}\right) \mid s_t\right]. \end{aligned}$$

## TWO THEORIES

- The only difference is the budget constraint.
- In the Barro (1979) case the government can only use one-period debt:

$$\tau_t + a_t = \frac{1}{1+r} a_{t+1} + g_t,$$

where $a_t$ is a risk free claim on time $t$ goods that the government purchased at time $t-1$; debt is $-a_t$.

- In the Lucas and Stokey (1983) case the government can use state-contingent debt:

$$\tau_t + a_{t-1}(s_t) = \int q_t(s_{t+1} \mid s_t)\, a_t(s_{t+1})\, ds_{t+1} + g_t,$$

where $a_{t-1}(s_t)$ is Arrow security that pays in state $s_t$ and $q_t(\cdot)$ are prices.

# TWO THEORIES

- To facilitate comparison we assume

$$q\left(s_{t+1} \mid s_t\right) = \frac{1}{1+r}\phi\left(s_{t+1} \mid s_t\right).$$

- This results in the budget constraint for the Lucas and Stokey (1983) case:

$$\tau_t + a_{t-1}\left(s_t\right) = \frac{1}{1+r}\mathbb{E}\left[a_{t+1} \mid s_t\right] + g_t,$$

where we define $\mathbb{E}\left[a_{t+1} \mid s_t\right] = \int a_t\left(s_{t+1}\right)\phi\left(s_{t+1} \mid s_t\right) ds_{t+1}$.

# EX-POST RETURNS

- Useful to distinguish between ex-ante and ex-post returns.
- Ex-post return on a portfolio of Arrow securities is

$$R_t\left(s_{t+1} \mid s_t\right) = \frac{a_t\left(s_{t+1}\right)}{\mathbb{E}\left[a_{t+1} \mid s_t\right]}$$

- The conditional expectation of the return on the government portfolio (ex-ante) is then

$$\mathbb{E}\left[R_t\left(s_{t+1} \mid s_t\right) \mid s_t\right] = 1 + r$$

# TWO THEORIES

- In Barro (1979) the optimal plan is

$$\tau_t = \frac{r}{1+r}\left[H\left(s_t\right) + a_t\right]$$

  and has the implication that taxes are a random walk

$$\mathbb{E}_t\left[\tau_{t+1}\right] = \tau_t.$$

- Government assets are also a random walk.

# TWO THEORIES

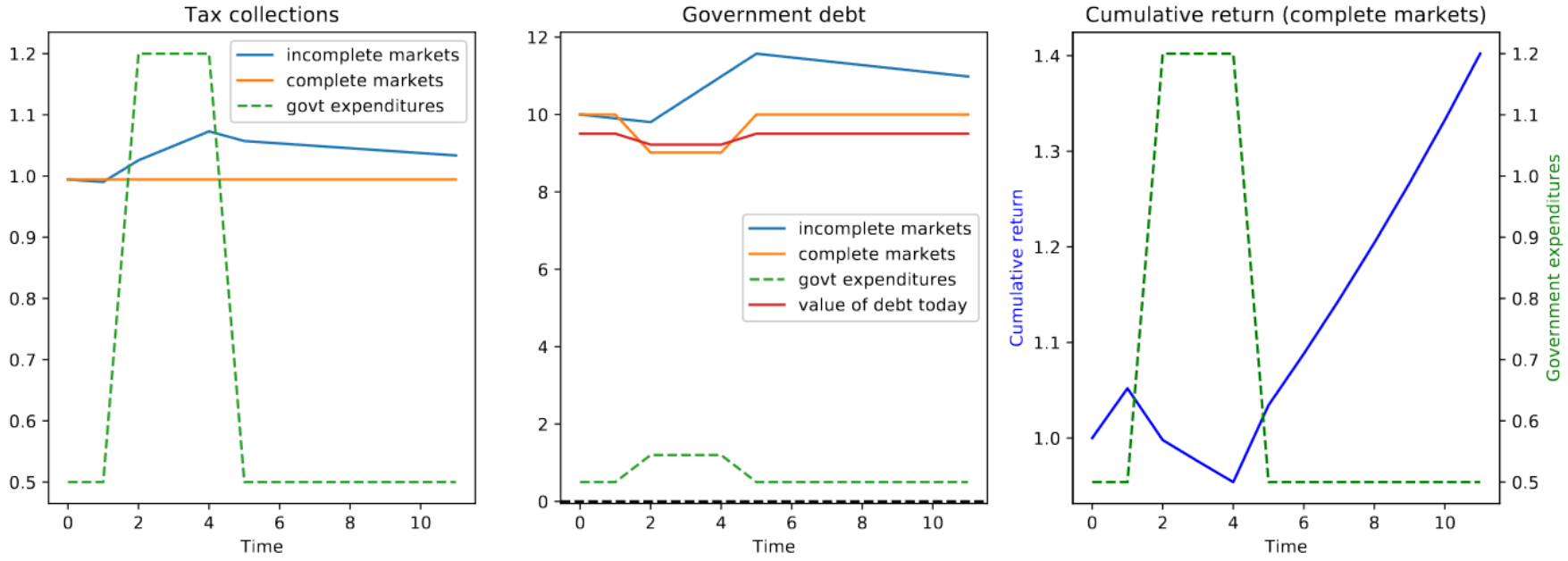
- This simplified version of Lucas and Stokey (1983) has the optimal plan that

$$\tau_t = \tau_0.$$

- The state-contingent asset-purchasing strategy supports this tax collection policy:

$$a_{t-1}\left(s_t\right) = H\left(s_t\right) - \frac{1+r}{r}\tau_0.$$

# TWO THEORIES




Tax and debt policy in complete and incomplete markets.

# GOVERNMENT BUDGET CONSTRAINT

# GOVERNMENT BUDGET CONSTRAINT

- Let $B_{t-1} = \sum_{j=1}^{n} B_{t-1}^{j}$ be the total nominal value of interest bearing government debt at $t-1$, where $B_{t-1}^{j}$ is the nominal value of zero coupon bonds of maturity $j$ at $t-1$.
- The government budget constraint at time $t$ is

$$B_t = B_{t-1} + r_{t-1,t} B_{t-1} + G_t - T_t - (M_t - M_{t-1}),$$

where $M_t$ is the nominal money supply and $r_{t-1,t}$ is implicitly defined by

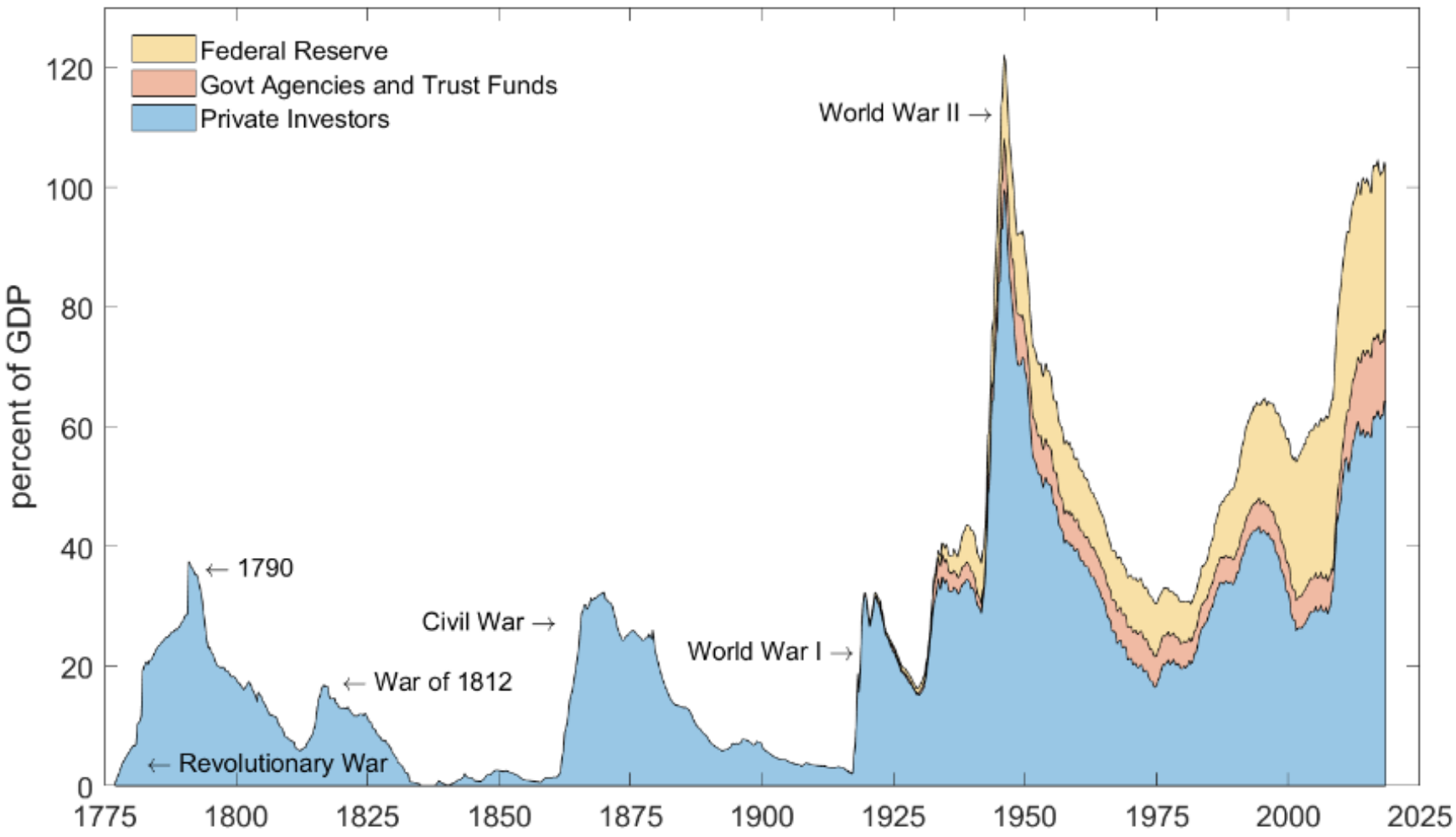
$$r_{t,t-1} B_{t-1} = \sum_{j=1}^{n} r_{t-1,t}^{j} B_{t-1}^{j}.$$

- All variables above are nominal.

# GOVERNMENT BUDGET CONSTRAINT

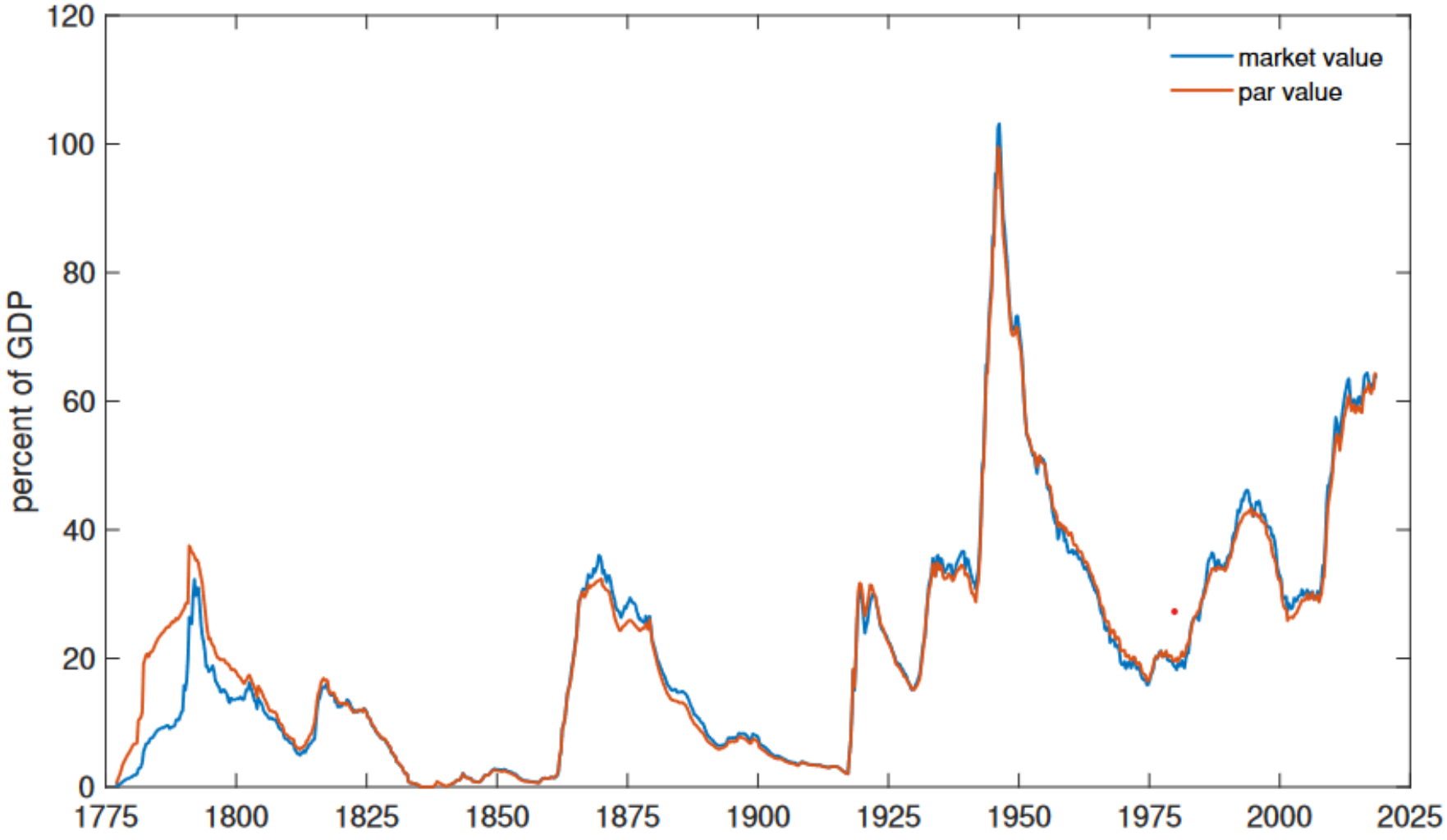
- Hall and Sargent make two adjustments to the U.S. Treasury's record of total public debt outstanding:
  1. net out holdings by the Federal Reserve and Government Agencies and Trust Funds;
  2. measure Treasury debt by its market value rather than its par value.

# GOVERNMENT BUDGET CONSTRAINT




Debt ownership.

# GOVERNMENT BUDGET CONSTRAINT




Debt value.

## GOVERNMENT BUDGET CONSTRAINT

- We can rewrite the budget constraint as

$$\begin{aligned}\frac{G_t}{Y_t} + r_{t-1,t}\frac{B_{t-1}}{Y_{t-1}} = \frac{T_t}{Y_t} &+ \left(\frac{B_t}{Y_t} - \frac{B_{t-1}}{Y_{t-1}}\right) + \frac{M_t - M_{t-1}}{Y_t} \\ &+ g_{t-1,t}\frac{B_{t-1}}{Y_{t-1}} + \pi_{t-1,t}\frac{B_{t-1}}{Y_{t-1}} \\ &+ r_{t-1,t}\left(g_{t-1,t} + \pi_{t-1,t}\right)\frac{B_{t-1}}{Y_{t-1}}\end{aligned}$$

  where $g_{t-1,t}$ denotes the growth rate of real GDP, and $\pi_{t-1,t}$ denotes the inflation rate.

- The first three terms on the right side record sources of government revenue as shares of GDP: taxes, new borrowing and money creation.
- The next two terms record the diminution of the debt/GDP ratio due to real GDP growth and inflation.

# GOVERNMENT BUDGET CONSTRAINT

- A "peacetime baseline" version of the constraint

$$
\begin{aligned}
\left(\frac{G}{Y}\right)^{\text{base}} + \left(r_{-1,0}\frac{B_{-1}}{Y_{-1}}\right)^{\text{base}} &= \left(\frac{T}{Y}\right)^{\text{base}} + \left(\frac{B}{Y} - \frac{B_{-1}}{Y_{-1}}\right)^{\text{base}} \\
&+ \left(\frac{M - M_{-1}}{Y}\right)^{\text{base}} \\
&+ \left(g_{-1,0}\frac{B_{-1}}{Y_{-1}}\right)^{\text{base}} + \left(\pi_{-1,0}\frac{B_{-1}}{Y_{-1}}\right)^{\text{base}} \\
&+ \left(r_{-1,t}\left(g_{-1,0} + \pi_{-1,0}\right)\frac{B_{-1}}{Y_{-1}}\right)^{\text{base}}
\end{aligned}
$$

# GOVERNMENT BUDGET CONSTRAINT

- In each period we can calculate deviation of "wartime" budget constraint from the "peacetime baseline".
- Hall and Sargent sum from the beginning of the war to the end of the war to get the total wartime deviation.

$$\sum_{t=T_1}^{T_2}\left[\frac{G_t}{Y_t}-\left(\frac{G}{Y}\right)^{\text{base}}\right]+\sum_{t=T_1}^{T_2}\left[r_{t-1,t}\frac{B_{t-1}}{Y_{t-1}}-\left(r_{-1,0}\frac{B_{-1}}{Y_{-1}}\right)^{\text{base}}\right]$$

$$=\sum_{t=T_1}^{T_2}\left[\frac{T_t}{Y_t}-\left(\frac{T}{Y}\right)^{\text{base}}\right]+\sum_{t=T_1}^{T_2}\left[\left(\frac{B_t}{Y_t}-\frac{B_{t-1}}{Y_{t-1}}\right)-\left(\frac{B}{Y}-\frac{B_{-1}}{Y_{-1}}\right)^{\text{base}}\right]$$

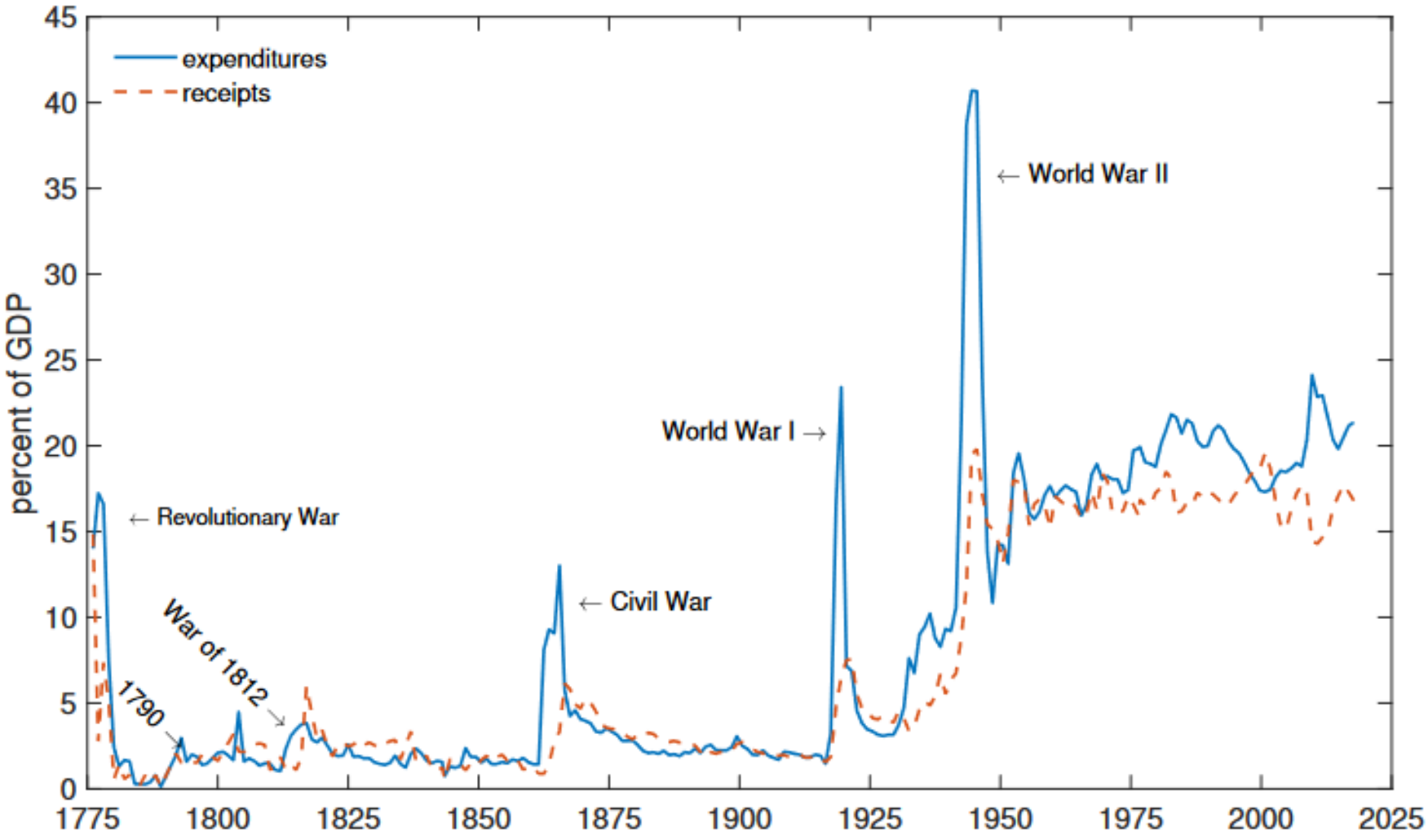
$$+\sum_{t=T_1}^{T_2}\left[\frac{M_t-M_{t-1}}{Y_t}-\left(\frac{M-M_{-1}}{Y}\right)^{\text{base}}\right]+\cdots$$

| War<br>Start - End<br>(U.S. entry - ) | (1)<br>government<br>spending | (2)<br>return<br>on debt | (3)<br>(1)+(2) | (4)<br>tax<br>revenue | (5)<br>debt<br>growth | (6)<br>money<br>growth | (7)<br>GDP<br>growth | (8)<br>inflation | (9)<br>cross<br>term | (10)<br>residual |
|---|---|---|---|---|---|---|---|---|---|---|
| War of 1812 | | | | | | | | | | |
| 1812:6 - 1815:2 | 7.34 | -0.20 | 7.14 | -2.35 | 10.60 | 0.00 | -0.16 | 0.06 | -0.39 | -0.62 |
| | | | | -32.9 | 148.5 | 0.0 | -2.2 | 0.8 | -5.5 | -8.7 |
| Mexican War | | | | | | | | | | |
| 1846:5 - 1848:2 | 2.26 | 0.20 | 2.47 | -0.06 | 2.72 | 0.00 | -0.06 | -0.01 | -0.00 | -0.12 |
| | | | | -2.4 | 110.4 | 0.0 | -2.5 | -0.5 | -0.1 | -4.8 |
| Civil War (Union) | | | | | | | | | | |
| 1861:4 - 1865:4 | 31.04 | 2.10 | 33.14 | 2.26 | 19.74 | 6.49 | 1.08 | 3.95 | 0.40 | -0.77 |
| | | | | 6.8 | 59.6 | 19.6 | 3.2 | 11.9 | 1.2 | -2.3 |
| Spanish-American War | | | | | | | | | | |
| 1898:4 - 1898:8 | 0.78 | 0.11 | 0.90 | 0.45 | -0.26 | 0.07 | 0.67 | 0.13 | 0.03 | -0.18 |
| | | | | 50.0 | -28.9 | 7.3 | 74.3 | 14.6 | 3.2 | -20.4 |
| World War I | | | | | | | | | | |
| 1914:7 - 1918:11 | 36.11 | 0.43 | 36.54 | 6.83 | 26.76 | 3.41 | 0.52 | 1.22 | 0.03 | -2.24 |
| | | | | 18.7 | 73.2 | 9.3 | 1.4 | 3.4 | 0.1 | -6.1 |
| (1917:4 -) | 36.93 | 0.30 | 37.23 | 7.76 | 27.79 | 2.59 | 0.05 | 0.76 | 0.00 | -1.73 |
| | | | | 20.8 | 74.6 | 7.0 | 0.1 | 2.1 | 0.0 | -4.6 |
| World War II | | | | | | | | | | |
| 1939:9 - 1945:8 | 129.50 | 0.10 | 129.60 | 49.91 | 54.78 | 11.32 | 15.42 | 9.62 | 0.26 | -11.71 |
| | | | | 38.5 | 42.3 | 8.7 | 11.9 | 7.4 | 0.2 | -9.0 |
| (1941:12 - ) | 116.48 | 2.00 | 118.48 | 35.80 | 54.53 | 11.96 | 8.99 | 6.05 | 0.43 | 0.71 |
| | | | | 30.2 | 46.0 | 10.1 | 7.6 | 5.1 | 0.4 | 0.6 |
| Korean War | | | | | | | | | | |
| 1950:6 - 1953:6 | 15.43 | -0.71 | 14.73 | 5.42 | 4.17 | 2.53 | 10.99 | -10.12 | 0.05 | 1.70 |
| | | | | 36.8 | 28.3 | 17.2 | 74.6 | -68.7 | 0.3 | 11.5 |
| Vietnam War | | | | | | | | | | |
| 1964:8 - 1973:6 | 5.53 | -2.13 | 3.41 | 1.39 | 0.44 | -0.60 | -5.55 | 3.91 | 0.19 | 3.63 |
| | | | | 40.8 | 12.9 | -17.8 | -163.0 | 114.9 | 5.7 | 106.5 |

# OPTIMAL RESPONSE

- Suppose $r = 0.06$, the net real interest rate in our model is 6%.
- Barro (1979) implies that a purely transitory increse in $g_t$ should be financed in 6% by taxes and in 94% by debt.
- We see it for the Civil War only.
- Caveat: were these increases in government purchases really transitory?

# RECEIPTS AND EXPENDITURES

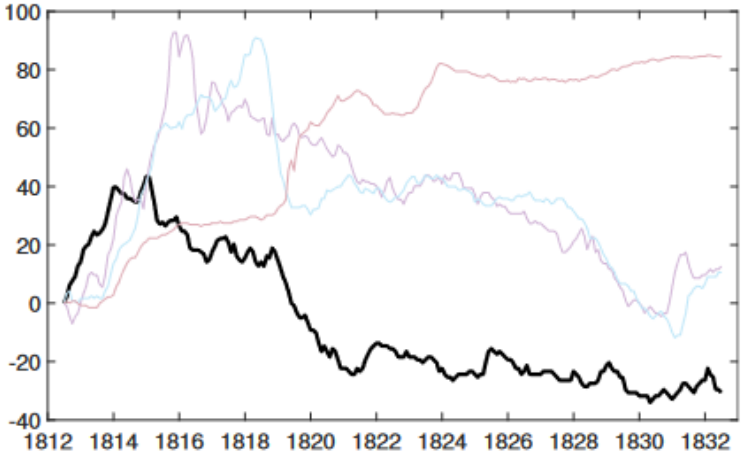


# OPTIMAL RESPONSE

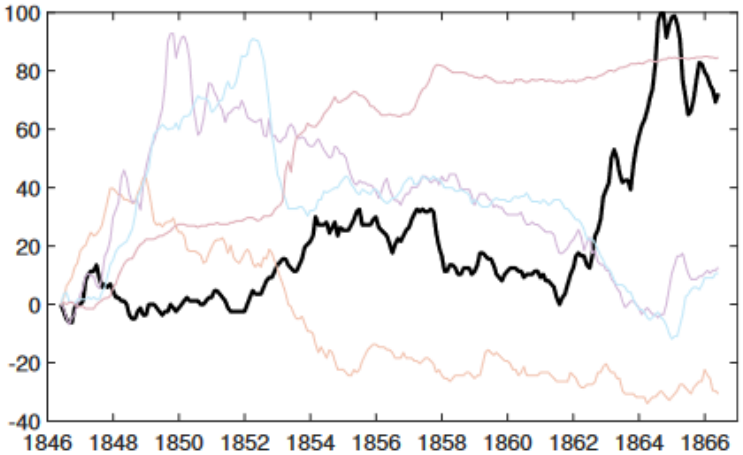
- Lucas and Stokey (1983) implies that unexpected increases in government spending should be absorbed by wartime decreases in returns to government creditors.
- With the exception of the Mexican War and the Korean War, the contribution of inflation is greater than the contribution of the nominal return on the debt.
- Negative wartime real returns.

| War<br>postwar period | 100 × Debt/GDP<br>(1)<br>end of war | (2)<br>15 years postwar | (3)<br>change | Contributions<br>(4)<br>nominal returns | (5)<br>real gdp growth | (6)<br>inflation | (7)<br>primary deficit | (8)<br>cross term | (9)<br>seignorage | (10)<br>residual |
|---|---|---|---|---|---|---|---|---|---|---|
| War for Independence | | | | | | | | | | |
| 1791-1806 | 33.3 | 9.6 | -23.8 | 11.3 | -15.4 | -7.6 | -17.3 | -1.0 | – | 6.3 |
| | | | | 47 | -65 | -32 | -73 | -4 | – | 26 |
| War of 1812 | | | | | | | | | | |
| 1815-1830 | 11.6 | 3.4 | -8.2 | 10.5 | -5.6 | 4.9 | -19.4 | 0.1 | – | 1.3 |
| | | | | 128 | -68 | 60 | -237 | 1 | – | 16 |
| Mexican War | | | | | | | | | | |
| 1848-1860† | 2.7 | 1.2 | -1.5 | 0.8 | -0.9 | -0.1 | -1.5 | -0.1 | – | 0.3 |
| | | | | 53 | -60 | -7 | -100 | -7 | – | 20 |
| Civil War (Union) | | | | | | | | | | |
| 1865-1880 | 22.1 | 15.6 | -6.5 | 21.4 | -14.5 | 13.5 | -29.5 | 0.1 | 1.2 | 1.3 |
| | | | | 329 | -223 | 208 | -454 | 2 | 18 | 20 |
| Spanish-American War | | | | | | | | | | |
| 1898-1913 | 4.6 | 2.2 | -2.4 | 0.9 | -1.2 | -1.1 | -1.9 | -0.1 | 0.8 | 0.1 |
| | | | | 38 | -50 | -46 | -79 | -4 | 33 | 4 |
| World War I | | | | | | | | | | |
| 1919-1929‡ | 28.6 | 20.2 | -8.4 | 12.5 | -6.4 | 2.4 | -20.3 | 0.3 | 2.0 | 1.0 |
| | | | | 149 | -76 | 29 | -242 | 4 | 24 | 12 |
| World War II | | | | | | | | | | |
| 1945-1960 | 90.1 | 35.7 | -54.4 | 14.3 | -15.8 | -38.9 | -13.0 | -0.6 | -0.3 | -0.2 |
| | | | | 26 | -29 | -71 | -24 | - 1 | -1 | 0 |
| Korean War | | | | | | | | | | |
| 1953-1968 | 49.9 | 21.8 | -28.1 | 14.0 | -20.3 | -10.8 | 4.0 | -0.8 | -5.8 | -8.5 |
| | | | | 50 | -72 | -38 | 14 | -3 | -21 | -30 |
| Vietnam War | | | | | | | | | | |
| 1973-1988 | 16.4 | 34.7 | +18.3 | 32.3 | -12.2 | -19.2 | 19.7 | -2.7 | -6.0 | 6.3 |
| | | | | 177 | -67 | -105 | 108 | -15 | -33 | 35 |

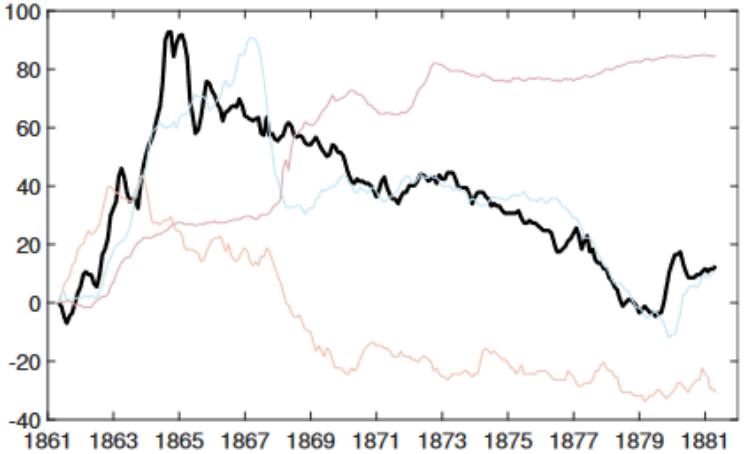
# PRICES



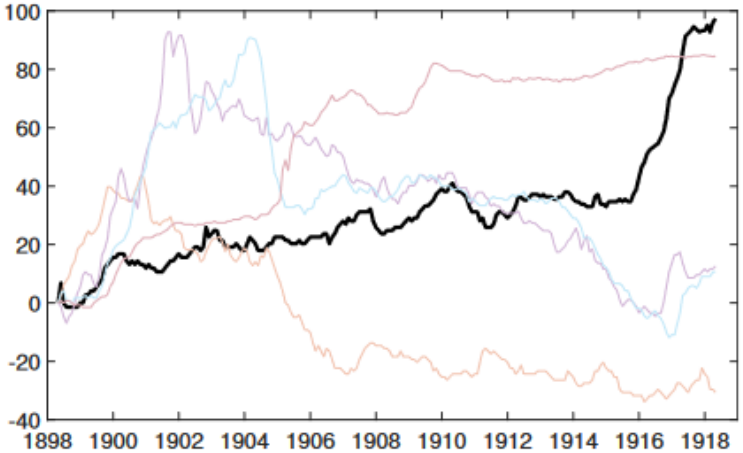

(a) War of 1812




(b) Mexican War



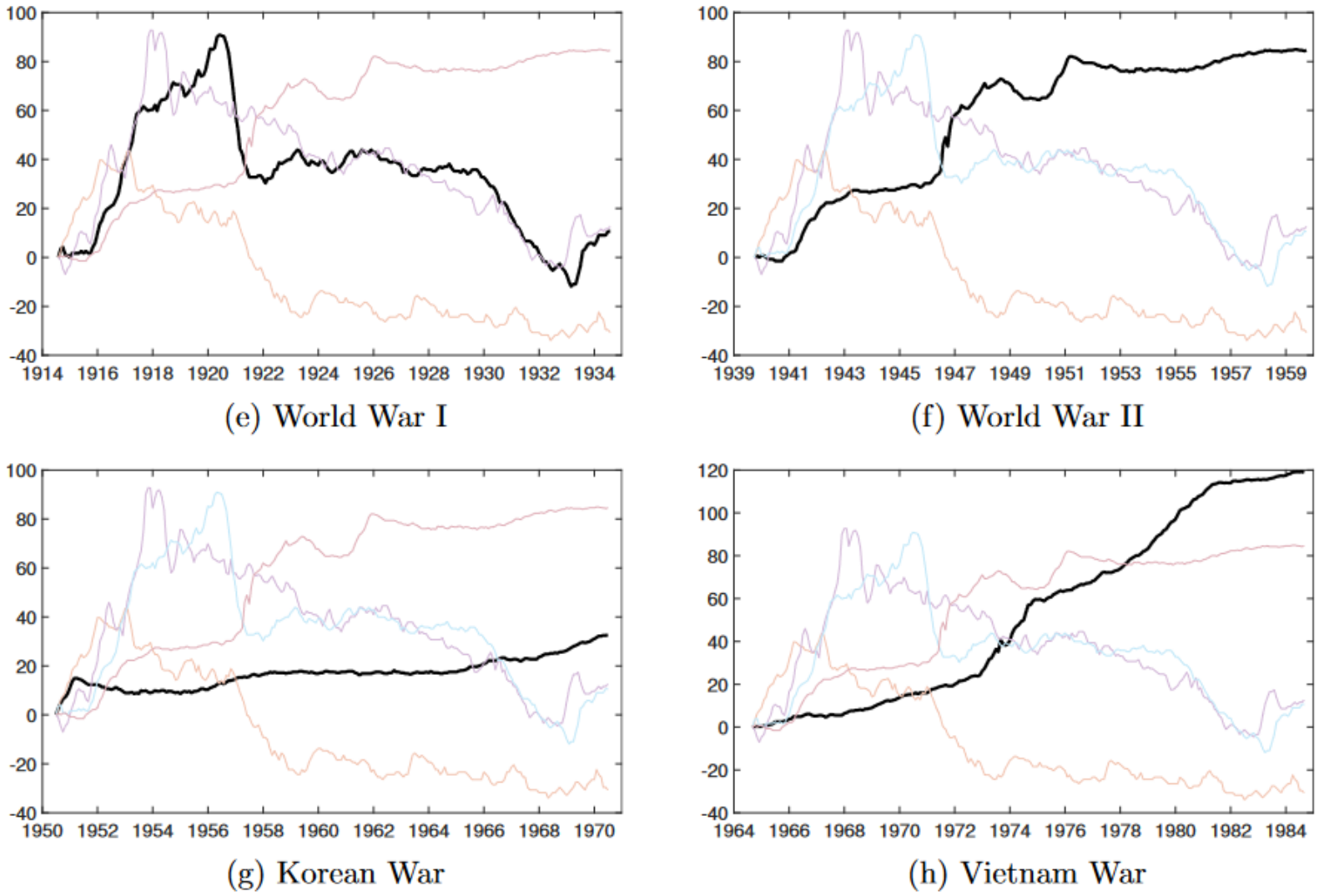

(c) Civil War




(d) Spanish-American War

Price level.

# PRICES

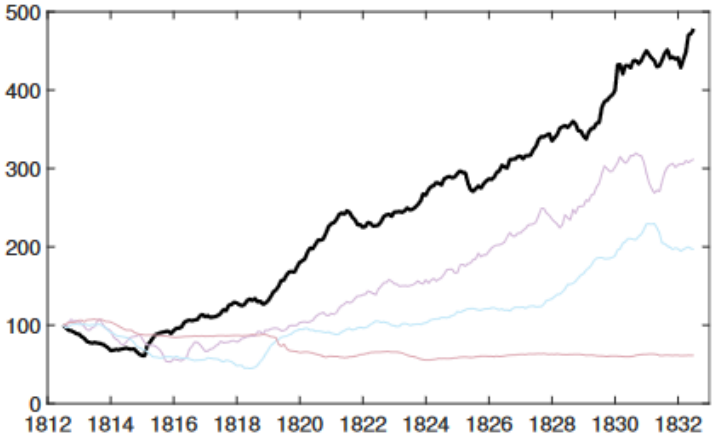



(e) World War I (f) World War II

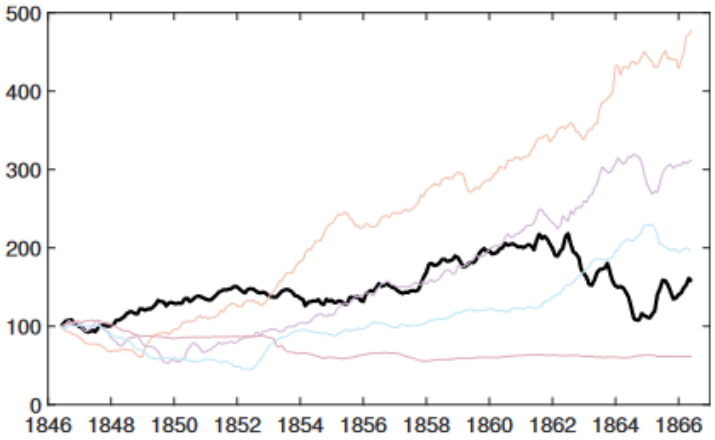
(g) Korean War (h) Vietnam War

Price level.

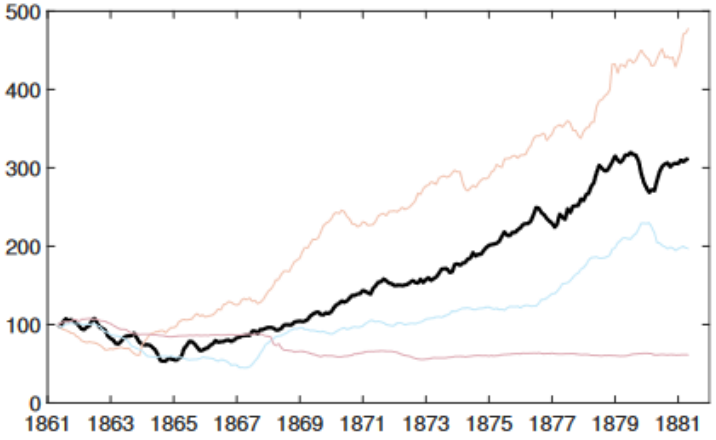
# RETURNS



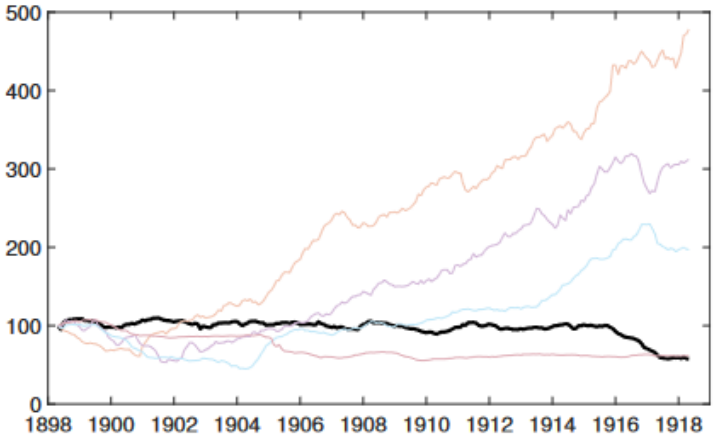

(a) War of 1812




(b) Mexican War



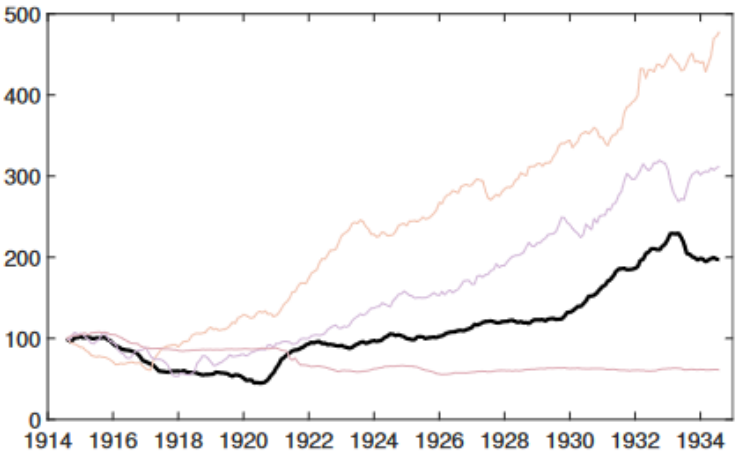

(c) Civil War



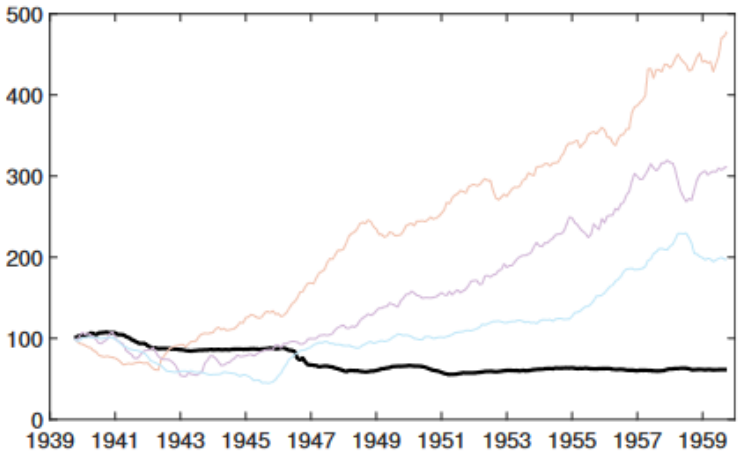

(d) Spanish-American War

Cumulative returns on debt portfolio.

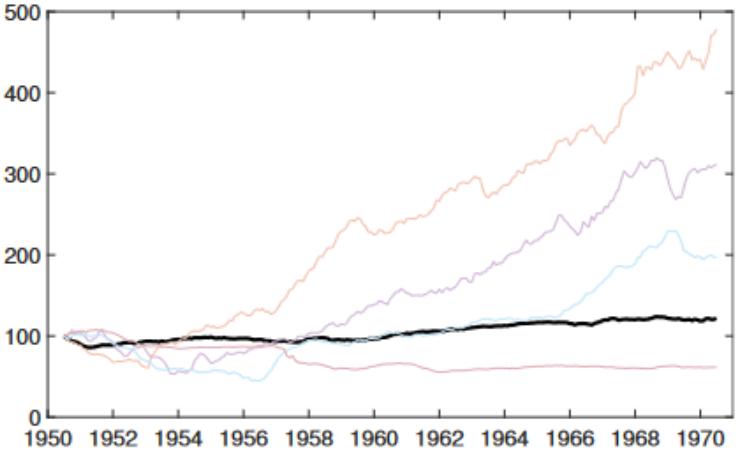
# RETURNS



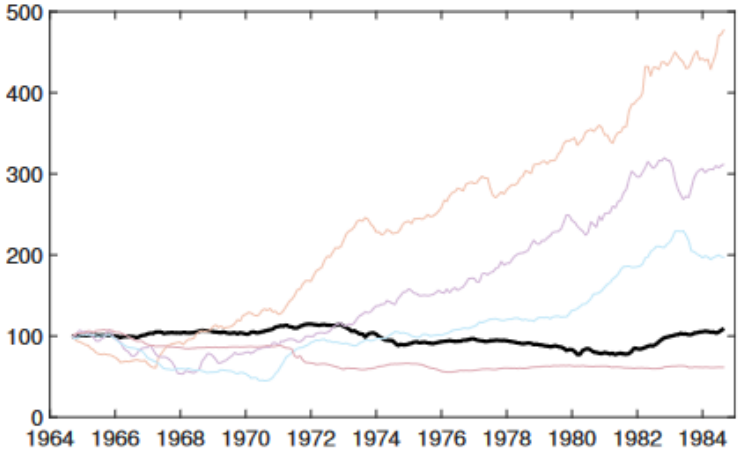

(e) World War I




(f) World War II




(g) Korean War




(h) Vietnam War

Cumulative returns on debt portfolio.

# CONCLUSIONS

- Evidence of Barro tax smoothing in most wars, but only during the Civil War the split between taxes and debt that the model recommends for a purely temporary expenditure surge.
- Negative wartime bond returns followed by positive postwar returns in the War of 1812, the Civil War, World War I and the Korean War as prescribed by the Lucas-Stokey model
- However, this model directs that bondholders should receive an immediate capital loss at the outbreak of a war. This happens only in the Korean War.
- The U.S. had little debt at the outbreak of most wars, the Lucas-Stokey action would not help the government's financial situation.